Hannah R. Kerner
Curriculum Vitae

School of Computing and Augmented Intelligence
Arizona State University, Tempe, AZ

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EDUCATION

Ph.D. School of Earth and Space Exploration, Arizona State University, 2019
B.S. Department of Computer Science, University of North Carolina at Chapel Hill, 2014

PROFESSIONAL APPOINTMENTS/EMPLOYMENT

Assistant Professor, School of Computing and Augmented Intelligence — 2022-Present
Center Faculty, Center for Global Conservation and Discovery Science — Tempe, AZ
Affiliate Faculty, School of Earth and Space Exploration
Affiliate Faculty, Swette Center for Sustainable Food Systems
Arizona State University

Chief Scientist — 2026-Present
Taylor Geospatial

Adjunct Assistant Professor — 2022-Present
Department of Geographical Sciences — College Park, MD
University of Maryland, College Park

Machine Learning/Artificial Intelligence Lead — 2020-Present
NASA Harvest and NASA Acres Consortia — College Park, MD

Assistant Research Professor — 2019-2022
Department of Geographical Sciences — College Park, MD
University of Maryland, College Park

Onboard Software Engineer — 2014-2015
Planet Labs (Planet, Inc.) — San Francisco, CA

HONORS AND AWARDS

2026 U.S. Awardee, Asia-Pacific Economic Cooperation (APEC) Science Prize for Innovation, Research, and Education (ASPIRE)
2025 Schmidt Sciences AI2050 Early Career Fellowship
2025 NSF CAREER award (CISE/III)
2024 Ira A. Fulton Schools of Engineering Best Teaching Award (Top 5%)
2023 ASU Students' Professor of Impact Award (×2)
2021 Top 10 of 100 projects solving problems related to the UN SDGs with AI, International Research Centre on Artificial Intelligence (IRCAI), for NASA Harvest
2021 Outstanding Research Faculty, Geographical Sciences (UMD)
2021 15 Leading Women in ML4EO, Radiant Earth Foundation
2021 Forbes 30 Under 30 in Science
2018 Google Women Techmakers Scholarship

GRANTS AND FELLOWSHIPS

2026 "Going Beyond the Fields of The World: Neural Fields and Features of The World" (PI)
Taylor Geospatial Innovation Fund, $1,202,191

2026 "Hawai'i Water AI (WAI): Transforming Reservoir Management through Dam Safety for Resiliency" (Co-I), NASA Water Resources Applications, $749,111

2026 "Promptable, Uncertainty-Aware GeoAI for In-Season Forecasting of Food Assistance Infrastructure in Conflict Zones" (Co-PI), Taylor Geospatial Institute Geospatial Innovation for Food Security Challenge (GIFS), $550,000

2026 "NAIRR: Tuning the Pretraining Data Hyperparameters: a data-centric approach to optimizing AI foundation models" (Co-PI), NSF National Artificial Intelligence Research Resource (NAIRR) Pilot Projects

2025 "Advancing AI for sustainable development with accessible and accountable geospatial foundation models" (PI), Schmidt Sciences AI2050 Early Career Fellowship, $500,000

2025 "AI for the Digital Planet – Advancing Machine Learning for Satellite Remote Sensing Data for the Future" (PI), NSF CAREER (CISE/III), $600,000

2025 "3D forest structure retrievals using AI and multi-modal satellite time-series data" (Co-I/ Institutional PI), NASA Decadal Survey Incubation Program: Science and Technology, $797,698

2025 "FARMGO: Enhancing Climate-Resilient Agriculture and ENSO Preparedness through NASA Data Integration and AI Geovisualization" (Co-I), NASA MUREP Earth Science Systems Research (MUREP ESSR), $400,000

2025 "Fields of The World: Phase 2" (PI), Taylor Geospatial Engine, $330,000

2024 "Agriculture Digitization Impact Assessment" (PI), Google Gift, $30,000

2024 "Sudan War Impact on Agriculture" (PI), NASA Harvest, $90,000

2024 "Enabling Next-Generation Data Science in Planetary Research with Machine Learning Dataset Creation Tools in JMARS" (PI)
NASA Planetary Data Archiving, Restoration, and Tools (PDART), $529,796

2024 "In Situ Exploration of Terrestrial Silica Sinter Deposits with Terra-Cam and Terra-TES" (Co-I)
NASA Planetary Science and Technology through Analog Research (PSTAR), $2,075,946

2024 "A Large-Scale Machine Learning Dataset to Unlock Global Field Boundary Segmentation" (PI)
Taylor Geospatial Engine, $100,000

2023 "Coordinating Regional and Global Condition Assessments in Support of Food Security" (Co-I)
NASA Goddard Space Flight Center, $70,000

2023 "A Data-Centric Approach to Improve Geographic Equity in Geospatial ML" (PI)
Google Research (Google Award for Inclusion Research program), $60,000

2023 "Lowering the Barriers to Planetary Science Studies with a Large Mars Model" (Co-PI)
NASA Jet Proposal Laboratory Strategic University Research Partnership (SURP), $60,000

2023 "RAPID: Rapid computational modeling of wildfires and management with emphasis on human activity" (Co-PI), NSF RAPID, $200,000

2023 "Machine Learning Datasets for Public Good with a Data-Centric AI Approach" (PI)
NASA Supplemental Open Source Software Awards, $149,892

2023 "Anomaly Visualization for Earth and Heliophysics GNSS Data using DORA" (Co-I)
NASA Multidomain Reusable Artificial Intelligence Tools, $202,922

2022 "NASA ACRES: A Climate Resilient Ecosystem Approach to Strengthening US Agriculture" (Co-I)
NASA Earth Science Applications: Agriculture, $15,000,000

2022 "NASA Harvest: NASA Food Security and Agriculture Consortium" (Co-I)
NASA, $15,000,000

2022 "EO-Enabled Regional and National Agricultural Monitoring in West Africa" (Co-I)
NASA SERVIR, $658,000

2022 "EO-Enabled Food Security Dashboard to Close Critical Data Gaps in Highly Food Insecure Maui County" (PI)
NASA Equity and Environmental Justice (EEJ), $250,000

2022 "Optimizing Crop Yield Data Collection for Supply Chain Enhancement" (Co-I)
Bill and Melinda Gates Foundation, $200,000

2021 "Planted Area Change Estimation for East Africa Virtual Crop Tour" (PI)
NASA Goddard Space Flight Center, $122,225

2021 "Earth Observations for Field Level Agricultural Resource Mapping (EO-FARM): Pilot in Rwanda in Support of NISR" (Co-I)
USDA Foreign Agricultural Service (FAS), $490,000

2021 "Strengthening Agri-Foods Data Systems to Inform Food Security Policies and Trade in Sub-Saharan Africa" (Co-I)
Alliance for a Green Revolution in Africa (AGRA), $546,859

2021 "Domain-Agnostic Outlier Detection in Science Data Sets" (PI)
NASA SMD ML/AI Cross-Divisional Use Case Demonstration, $110,475

2020 "Quantifying volcanic activity from space with multiple sensors: The CEOS volcano demonstrator project" (Co-I/Institutional PI)
NASA Earth Surface and Interior (ESI), $1,156,857

2020 "In-Season Crop Monitoring Using Earth Observation in Major Food-Producing Countries to Mitigate Market Uncertainty Caused by COVID-19 Pandemic" (PI)
NASA Rapid Response and Novel Research in Earth Sciences, $100,000

2020 Microsoft AI for Earth Grant (PI), $10,000 Azure credits

2020 "Earth Observations for Field Level Agricultural Resource Mapping (EO-Farm): Pilot in Kenya and Mexico in Support of Small Holders" (Co-I)
SwissRe Foundation, $750,000

2019 "Novelty-Driven Onboard Targeting for MSL and Mars 2020 Rovers" (Co-I)
NASA Center Innovation Fund Advanced Concepts, $200,000

PUBLICATIONS

Preprints Under Review

1. Corley, I., Rao, A., Rolf, E., Klemmer, K., Shelhamer, E., Lehmann, N., Russwurm, M., Mai, G., Jacobs, N., **Kerner, H.** (2026). MIND the Gap: A Geographic Implicit Neural Representation with Adjustable Spatial Scale. ArXiv preprint, `https://arxiv.org/abs/2609.25454`.

2. Tokas, B., Nair, R., Jackson, R., **Kerner, H.** (2026). Hide and Seek: HIDE Concepts to SEEK explanations for black box models. ArXiv preprint, `https://arxiv.org/abs/2610.13935`.

3. Doerksen, K. and **Kerner, H.** (2026). EarthShift: a benchmark for measuring robustness to real-world distribution shifts in Earth observation. ArXiv preprint, `https://arxiv.org/abs/2605.29330`.

Peer-Reviewed Conference Proceedings

4. Doerksen, K., **Kerner, H.**, Ferraz, A., Lu, S., Tseng, G., Berndt, S. (2027). Rooted in embeddings: Evaluating geospatial foundation model representations for forest canopy height estimation. To appear in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops*, 4th Workshop on Computer Vision for Earth Observation.
5. Hess-Dunlop, A., Zvonkov, I., Tárano, A. M., Latimer, M., Loque, C., Wirth, M., DeSilva, L., Chan Hodges, S., Whitcraft, A., Beery, S., **Kerner, H.** (2027). Deer AI: Supporting Invasive Axis Deer Mitigation with Automated Detection from Aerial Videos in Maui County, Hawaii. To appear in *Proceedings of the Thirty-Ninth Annual Conference on Innovative Applications of Artificial Intelligence (IAAI-27)*. 🎤 Oral presentation
6. Perasso, B., Ciranni, M., Tokas, B., Nair, R., **Kerner, H.**, Pastore, V. P. (2027). Beyond the Generative Trilemma: Bias Amplification in Diffusion-GAN Hybrids. To appear in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*.
7. Corley, I., Robinson, C., Marcus, J., and **Kerner, H.** (2027). Fields of the Planet: Field Boundary Mapping Beyond 10m. To appear in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, `https://arxiv.org/abs/2607.04449`.
8. Corley, I. and Lehmann, N. and Robinson, C. and Tseng, G. and Fuller, A. and Alemohammad, H. and Shelhamer, E. and Marcus, J. and **Kerner, H.** (2026). No One Knows the State of the Art in Geospatial Foundation Models. To appear in *Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS)*, Position Paper Track, `https://arxiv.org/abs/2605.12678`.
9. Nair, R., Kumar, S., **Kerner, H.** (2026). A Large-scale Evaluation of Text-guided Models for Facial Editing. *Proceedings of the 34th ACM International Conference on Multimedia*, `https://arxiv.org/abs/2608.28802`. 🎤 Oral presentation
10. Kaur, A., Purohit, M., Muhawenayo, G., Rolf, E., **Kerner, H.** (2026). Pretrain Where? Investigating How Pretraining Data Diversity Impacts Geospatial Foundation Model Performance. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, EarthVision Workshop, `https://arxiv.org/abs/2604.21104`.
11. Cui, C., Jackson, R., Herrera, B., Tárano, A. M., **Kerner, H.** (2026). Multi-Head LatentMoE and Head Parallel: Communication-Efficient and Deterministic MoE Parallelism. *Proceedings of the International Conference on Machine Learning (ICML)*, `https://arxiv.org/abs/2602.04870`.
12. Muhawenayo, G., Robinson, C., Khanal, S., Fang, Z., Corley, I., Wollam, A., Gao, T., Strnad, L., Avery, R., Estes, L., Tárano, A. M., Jacobs, N., and **Kerner, H.** (2026). PRUE: A Practical Recipe for Field Boundary Segmentation at Scale. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, `https://arxiv.org/abs/2603.27101`.
13. Purohit, M., Gajera, B., Mehta, I., Tokas, B., Adler, J., Lu, S., Dickenshied, S., Diniega, S., Bue, B., Rebbapragada, U., and **Kerner, H.** (2026). MOMO: Mars Orbital MOdel Foundation Model for Mars Orbital Applications. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, `https://arxiv.org/abs/2604.02719`.
14. Herzog, H., Bastani, F., Zhang, Y., Tseng, G., Redmon, J., Sablon, H., Park, R., Morrison, J., Buraczynski, A., Farley, K., Hansen, J., Howe, A., Johnson, P. A., Otterlee, M., Schmitt,

T., Pitelka, H., Daspit, S., Ratner, R., Wilhelm, C., Wood, S., Jacobi, M., **Kerner, H.**, Shelhamer, E., Farhadi, A., Krishna, R., and Beukema, P. (2026). OlmoEarth: Stable Latent Image Modeling for Multimodal Earth Observation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, https://arxiv.org/abs/2511.13655.

15. Nair, R., Tokas, B., and **Kerner, H.** (2026). A Woman with a Knife or A Knife with a Woman? Measuring Directional Bias Amplification in Image Captions. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, https://arxiv.org/abs/2503.07878.
16. Purohit, M., Gajera, B., Malaviya, V., Mehta, I., Kasodekar, K., Adler, J., Lu, S., Rebbapragada, U., and **Kerner, H.** (2025). Mars-Bench: A Benchmark for Evaluating Foundation Models for Mars Science Tasks. In *Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS)*, Datasets and Benchmarks Track, https://arxiv.org/abs/2510.24010.
17. Tokas, B., Nair, R., **Kerner, H.** (2025). DPA: A one-stop metric to measure bias amplification in classification datasets. In *Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS)*, https://arxiv.org/abs/2412.11060.
18. Butsko, C., Tseng, G., Van Tricht, K., Milli, G., Rolnick, D., Cartuyvels, R., Becker Reshef, I., Szantoi, Z., **Kerner, H.** (2025). Deploying geospatial foundation models in the real world. In *Proceedings of The TerraBytes ICML Workshop: Towards global datasets and models for Earth Observation*, PMLR 292:13-31, https://proceedings.mlr.press/v292/butsko25a.html.
19. Tseng, G., Fuller, A., Reil, M., Herzog, H., Beukema, P., Bastani, F., Green, J. R., Shelhamer, E., **Kerner, H.**, and Rolnick, D. (2025). Galileo: Learning Global and Local Features in Pretrained Remote Sensing Models. In *Proceedings of the 42nd International Conference on Machine Learning*, https://arxiv.org/abs/2502.09356.
20. Nair, R., Tseng, G., Rolf, E., Tokas, B., and **Kerner, H.** (2025). Classification Drives Geographic Bias in Street Scene Segmentation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, Workshop on Fair, Data-Efficient, and Trusted Computer Vision, https://arxiv.org/abs/2412.11061.
21. Tseng, G., **Kerner, H.**, and Rolnick, D. (2025). Task-Informed Meta-Learning for Remote Sensing. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, EarthVision Workshop, https://openaccess.thecvf.com/content/CVPR2025W/EarthVision/html/Tseng_Task-Informed_Meta-Learning_for_Remote_Sensing_CVPRW_2025_paper.html.
22. Mohan, A., Gomez-Patron, A., Pritchard, M., **Kerner, H.** (2025). Hotspotter: A Generalizable Pipeline for Automated Detection of Subtle Volcanic Thermal Features in Satellite Images. *Proceedings of the AAAI Conference on Artificial Intelligence, 39(28)*, pp. 28857-28863, https://doi.org/10.1609/aaai.v39i28.35151.
23. Hopkins, L., Wong, W-K., **Kerner, H.**, Li, F., Hutchinson, R. (2025). Data Augmentation Approaches for Satellite Imagery. *Proceedings of the AAAI Conference on Artificial Intelligence, 39(27)*, pp. 28097-28105, https://doi.org/10.1609/aaai.v39i27.35028.
24. **Kerner, H.**, Chaudhari, S., Ghosh, A., Robinson, C., Ahmad, A., Choi, E., Jacobs, N., Holmes, C., Mohr, M., Dodhia, R., Lavista Ferres, J. M., Marcus, J. (2025). Fields of The World: A Machine Learning Benchmark Dataset For Global Agricultural Field Boundary

Segmentation. *Proceedings of the AAAI Conference on Artificial Intelligence, 39(27)*, pp. 28151-28159, https://arxiv.org/abs/2409.16252.

25. Cui, C., Yan, Z., Muhawenayo, G., and **Kerner, H.** (2024). An All-MLP Sequence Modeling Architecture That Excels at Copying. *International Conference on Machine Learning (ICML) Workshops*, Next Generation of Sequence Modeling Architectures Workshop, https://arxiv.org/abs/2406.16168v1.
26. Dhakal, A., Ahmad, A., Khanal, S., Sastry, S., **Kerner, H.**, and Jacobs, N. (2024). Sat2Cap: Mapping Fine-Grained Textual Descriptions from Satellite Images. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, EarthVision 2024, pp. 533-542, https://openaccess.thecvf.com/content/CVPR2024W/EarthVision/papers/Dhakal_Sat2Cap_Mapping_Fine-Grained_Textual_Descriptions_from_Satellite_Images_CVPRW_2024_paper.pdf. 🏅 Best Paper Award
27. Rolnick, D., Aspuru-Guzik, A., Beery, S., Dilkina, B., Donti, P. L., Ghassemi, M., **Kerner, H.**, Monteleoni, C., Rolf, E., Tambe, M., White, A. (2024). Position Paper: Application-Driven Innovation in Machine Learning. In *Proceedings of the International Conference on Machine Learning (ICML)*, https://arxiv.org/abs/2403.17381.
28. Rolf, E., Klemmer, K., Robinson, C., and **Kerner, H.** (2024). Position Paper: Mission Critical–Satellite Data is a Distinct Modality in Machine Learning, In *Proceedings of the International Conference on Machine Learning (ICML)*, https://arxiv.org/abs/2402.01444. 🌟 Spotlight (3.5% acceptance rate)
29. Purohit, M., Adler, J., and **Kerner, H.** (2024). ConeQuest: A Benchmark for Cone Detection on Mars. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 6026-6035, https://openaccess.thecvf.com/content/WACV2024/html/Purohit_ConeQuest_A_Benchmark_for_Cone_Segmentation_on_Mars_WACV_2024_paper.html. 🎤 Oral presentation (2.5% acceptance rate)
30. Malvi, S., Shah, H., Chandarana, N., Purohit, M., Adler, J., and **Kerner, H.** (2023). Automated Multi-class Crater Segmentation in Mars Orbital Images. In *Proceedings of the 31st International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*, pp. 110-120, https://dl.acm.org/doi/abs/10.1145/3615886.3627748.
31. Lacoste, A., Lehmann, N., Rodriguez, P., Sherwin, E. D., **Kerner, H.**, Lütjens, B., Irvin, J. A., Dao, D., Alemohammad, H., Drouin, A., Gunturkun, M., Huang, G., Vazquez, D., Newman, D., Bengio, Y., Ermon, S., Zhu, X. (2023). GEO-Bench: Toward Foundation Models for Earth Monitoring. In *Proceedings of the Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, https://arxiv.org/abs/2306.03831.
32. Zvonkov, I., Tseng, G., Nakalembe, C., and **Kerner, H.** (2023). OpenMapFlow: A Library for Rapid Map Creation with Machine Learning and Remote Sensing Data. In *Proceedings of the AAAI Conference on Artificial Intelligence, 37(12)*, 14655-14663, https://doi.org/10.1609/aaai.v37i12.26713.
33. Manimurugan, S., Singaram, R., Nakalembe, C., and **Kerner, H.** (2022). Geo-referencing crop labels from street-level images using Structure from Motion. In *Proceedings of the 73rd International Astronautical Congress (IAC)*.
34. **Kerner, H. R.** and Adler, J. B. (2022). Guiding Field Exploration on Earth and Mars with Outlier Detection. In *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, https://doi.org/10.1109/IGARSS46834.2022.9884366.

35. Tseng, G., Zvonkov, I., Nakalembe, C., **Kerner, H.** (2021). CropHarvest: a global satellite dataset for crop type classification. *Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, https://openreview.net/forum?id=JtjzUXPEaCu.

36. Tseng, G., **Kerner, H.**, Nakalembe, C., and Becker-Reshef, I. (2021). Learning to predict crop type from heterogeneous sparse labels using meta-learning. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, EarthVision 2021, https://openaccess.thecvf.com/content/CVPR2021W/EarthVision/papers/Tseng_Learning_To_Predict_Crop_Type_From_Heterogeneous_Sparse_Labels_Using_CVPRW_2021_paper.pdf.

37. Horton, P., **Kerner, H.**, Jacobs, S., Cisneros, E., Wagstaff, K. L., and Bell III, J. F. (2021). Integrating Novelty Detection Capabilities with MSL Mastcam Operations to Enhance Data Analysis. IEEE Aerospace Conference, Big Sky, Montana, March 6-13, https://arxiv.org/abs/2103.12815.

38. Wagstaff, K. L., Francis, R., **Kerner, H.**, Lu, S., Nerrise, F. (2020). Novelty-Driven Onboard Targeting for Mars Rovers. *International Symposium on Artificial Intelligence, Robotics and Automation in Space (i-SAIRAS)*, https://www.hou.usra.edu/meetings/isairas2020fullpapers/pdf/5056.pdf.

39. **Kerner, H. R.**, Wellington, D. F., Wagstaff, K. L., Bell III, J. F., Kwan, C., Ben Amor, H. (2019). Novelty Detection for Multispectral Images with Application to Planetary Exploration. *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 9484-9491, https://doi.org/10.1609/aaai.v33i01.33019484. 🎤 Oral presentation

Peer-Reviewed Journal Articles

40. Simumba, N., Lehmann, N., Fraccaro, P., Alemohammad, H., De Mel, G., Khan, S., Maskey, M., Longepe, N., Zhu, X. X., **Kerner, H.**, Bernabe-Moreno, J., and Lacoste, A. (2025). GEO-Bench-2: From Performance to Capability, Rethinking Evaluation in Geospatial AI. *Transactions on Machine Learning Research*, https://arxiv.org/abs/2511.15658.

41. Klemmer, K., Rolf, E., Russwurm, M., Camps-Valls, G., Czerkawski, M., Ermon, S., Francis, A., Jacobs, N., **Kerner, H.** R., Mackey, L., Mai, G., Mac Aodha, O., Reichstein, M., Robinson, C., Rolnick, D., Shelhamer, E., Sitzmann, V., Tuia, D., Zhu, X. (2026). Earth Embeddings: Towards AI-centric Representations of our Planet. *IEEE Geoscience and Remote Sensing Magazine*, https://doi.org/10.1109/MGRS.2026.3710416.

42. Adebayo, A., Nakalembe, C., Skakun, S., Botchway, D., Ginsburg, A., Gadhiya, Y. G., Adjei, M., **Kerner, H.**, Becker-Reshef, I. (2026). Optimizing Satellite-Based Cropland Area Estimation Through Integrated Map Accuracy Assessment and Stratified Sampling Design Across Six African Countries. *International Journal of Applied Earth Observation and Geoinformation, 148*, p. 105223, https://doi.org/10.1016/j.jag.2026.105223.

43. Zhang, C., **Kerner, H.**, Wang, S., Hao, P., Li, Z., Hunt, K.A., Abernethy, J., Zhao, H., Gao, F., Di, L. and Guo, C. (2025). Remote sensing for crop mapping: A perspective on current and future crop-specific land cover data products. *Remote Sensing of Environment, 330*, p. 114995, https://doi.org/10.1016/j.rse.2025.114995.

44. Nakalembe, C., Zvonkov, I., **Kerner, H.**, Frimpong, D. B., Mwangi, K., Kioko, J., Tokas, B., Jawanjal, K., Smith, I. A., Paliyam, A., Wakhanala, C. A., Tárano, A. M., Jha, S., and Loh, P. M. (2025). Helmets Labeling Crops: Kenya Crop Type Dataset Created via

Helmet-Mounted Cameras and Deep Learning. *Scientific Data, 12(1)*, 1496, https://www.nature.com/articles/s41597-025-05762-7.

45. Nakalembe, C., **Kerner, H.**, Zvonkov, I., Humber, M., Galvez, A., Makabe, E., Venturini, S., Becker-Reshef, I. (2025). A Framework for EO-Based National Agricultural Monitoring (EO-NAM)- For the African Context. *npj Sustainable Agriculture, 3(45)*, https://www.nature.com/articles/s44264-025-00083-z.
46. Rufin, P., et al. (**Kerner, H.** one of 16 authors) (2025). Accelerating research on SDG2 'Zero Hunger' by opening commercial very-high resolution satellite image archives. *Proceedings of the National Academy of Sciences, 122 (7)*, e2410246122, https://www.pnas.org/doi/10.1073/pnas.2410246122.
47. Nakalembe, C. L., Frimpong, D. B., **Kerner, H. R.**, Sarr, M. A. (2025). A 40-Year Remote Sensing Analysis of Spatiotemporal Temperature and Rainfall Patterns in Senegal. *Frontiers in Climate, 7*, https://www.frontiersin.org/journals/climate/articles/10.3389/fclim.2025.1462626/full.
48. Peng, J., Li, J., Ingalls, T. C., Schill, S. R., **Kerner, H. R.**, and Asner, G. P. (2025). A novel deep learning algorithm for broad scale seagrass extent mapping in shallow coastal environments. *ISPRS Journal of Photogrammetry and Remote Sensing, 220*, 277-294, https://doi.org/10.1016/j.isprsjprs.2024.12.008.
49. Gray, P. C., Boss, E., Prochaska, J. X., **Kerner, H.**, Demeaux, C. B., and Lehahn, Y. (2024). The Promise and Pitfalls of Machine Learning in Ocean Remote Sensing. *Oceanography, 37(3)*, pp. 52-63, https://doi.org/10.5670/oceanog.2024.511.
50. **Kerner, H.**, Nakalembe, C., Yeh, B., Zvonkov, I., Skakun, S., Becker-Reshef, I., and McNally, A. (2024). Satellite Data Shows Resilience of Tigrayan Farmers in Crop Cultivation During Civil War. *Science of Remote Sensing*, 10, 100140, https://www.sciencedirect.com/science/article/pii/S2666017224000245?via%3Dihub.
51. **Kerner, H.**, Nakalembe, C., Yang, A., Zvonkov, I., McWeeny, R., Tseng, G., and Becker-Reshef, I. (2024). How accurate are existing land cover maps for agriculture in Sub-Saharan Africa? *Nature Scientific Data, 11(486)*, https://www.nature.com/articles/s41597-024-03306-z.
52. Kebede, E. A., Abou Ali, H., Clavelle, T., Froehlich, H. E., Gephart, J. A., Hartman, S., Herrero, M., **Kerner, H.**, Mehta, P., Nakalembe, C., Ray, D. K., Siebert, S., Thornton, P., Davis, K. F. (2024). Assessing and addressing the global state of food production data scarcity. *Nature Reviews Earth & Environment*, p. 1-17, https://www.nature.com/articles/s43017-024-00516-2.
53. Prieur, N. C., Amaro, B., Gonzalez, E., **Kerner, H.**, Medvedev, S., Rubanenko, L., Werner, S., Xiao, Z., Zastrozhnov, D., and Lapôtre, M. G. (2023). Automatic Characterization of Boulders on Planetary Surfaces From High-Resolution Satellite Images. *Journal of Geophysical Research: Planets, 128(11)*, e2023JE008013, https://doi.org/10.1029/2023JE008013. 🏆 Top 10% Viewed Article in 2023
54. Nakalembe, C. and **Kerner, H.** (2023). Considerations for AI-EO for agriculture in Sub-Saharan Africa. *Environmental Research Letters, 18(4)*, https://iopscience.iop.org/article/10.1088/1748-9326/acc476.
55. Manheim, M. R., Henriksen, M. R., Robinson, M. S., **Kerner, H. R.**, Karas, B. A., Becker, K. J., Chojnacki, M., Sutton, S. S., Blewett, D. T. (2022). High-Resolution Regional Digital

Elevation Models and Derived Products from MESSENGER MDIS Images. *Remote Sensing, 14*, 3564, https://doi.org/10.3390/rs14153564.

56. Rice, M.S., Seeger, C., Bell, J., Calef, F., St Clair, M., Eng, A., Fraeman, A.A., Hughes, C., Horgan, B., Jacob, S., Johnson, J., **Kerner, H.**, Kinch, K., Lemmon, M., Million, C., Starr, M., and Wellington, D. (2022). Spectral diversity of rocks and soils in Mastcam observations along the Curiosity rover's traverse in Gale crater, Mars. *Journal of Geophysical Research: Planets*, p.e2021JE007134, https://doi.org/10.1029/2021JE007134.

57. **Kerner, H. R.**, Sahajpal, R., Pai, D. B., Skakun, S., Puricelli, E., Hosseini, M., Meyer, S., and Becker-Reshef, I. (2022). Phenological normalization can improve in-season classification of maize and soybean: A case study in the central US Corn Belt. *Science of Remote Sensing, 6*, 100059, https://doi.org/10.1016/j.srs.2022.100059.

58. **Kerner, H. R.**, Rebbapragada, U., Wagstaff, K. L., Lu, S., Dubayah, B., Huff, E., Raman, V., and Kulshrestha, S. (2022). Domain-Agnostic Outlier Ranking Algorithms—A Configurable Pipeline for Facilitating Outlier Detection in Scientific Datasets. *Frontiers in Astronomy and Space Sciences, 9*, 867947, https://doi.org/10.3389/fspas.2022.867947.

59. Handwerger, A. L., Jones, S. Y., Amatya, P., **Kerner, H. R.**, Kirschbaum, D. B., and Huang, M. H. (2021). Strategies for landslide detection using open-access synthetic aperture radar backscatter change in Google Earth Engine. *Natural Hazards and Earth System Sciences Discussions*, 22, pp. 753-773, https://doi.org/10.5194/nhess-22-753-2022.

60. Shirzaei, M., Khoshmanesh, M., Ojha, C., Werth, S., **Kerner, H.**, Carlson, G., Sherpa, S. F., Zhai, G., and Lee, J. (2021). Persistent impact of spring floods on crop loss in U.S. Midwest. *Weather and Climate Extremes, 34*, 100392, https://doi.org/10.1016/j.wace.2021.100392.

61. Gray, P. C., Chamorro, D. F., Ridge, J. T., **Kerner, H. R.**, Ury, E. A., and Johnston, D. W. (2021). Temporally Generalizable Land Cover Classification: A Recurrent Convolutional Neural Network Unveils Major Coastal Change through Time. *Remote Sensing, 13(19)*, 3953, https://doi.org/10.3390/rs13193953.

62. Lawal, A., **Kerner, H.**, Becker-Reshef, I., Meyer, S. (2021). Mapping the Location and Extent of 2019 Prevent Planting Acres in South Dakota Using Remote Sensing Techniques. *Remote Sensing, 13(13)*, 2430, https://www.mdpi.com/2072-4292/13/13/2430.

63. Hosseini, M., **Kerner, H.**, Sahajpal, R., Puricelli, E., Lu, Y-H., Lawal, A., Humber, M. L., Mitkish, M., Meyer, S., Becker-Reshef, I. (2020). Evaluating the Impact of the 2020 Iowa Derecho on Corn and Soybean Fields Using Synthetic Aperture Radar. *Remote Sensing, 12(23)*, 3878, https://www.mdpi.com/2072-4292/12/23/3878.

64. **Kerner, H. R.**, Wagstaff, K. L., Bue, B. D., Wellington, D. F., Jacob, S., Horton, P., Bell, J. F., Kwan, C., Ben Amor, H. (2020). Comparison of Novelty Detection Methods for Multispectral Images in Rover-Based Planetary Exploration Missions. *Data Mining and Knowledge Discovery, 34,* pp. 1642–1675, https://doi.org/10.1007/s10618-020-00697-6.

65. **Kerner, H. R.**, Hardgrove, C., Czarnecki, S., Gabriel, T. S. J., Mitrofanov, I., Litvak, M., Sanin, A., Lisov, D. (2020). Analysis of Active Neutron Measurements from the Mars Science Laboratory Dynamic Albedo of Neutrons Instrument: Intrinsic Variability, Outliers, and Implications for Future Investigations. *Journal of Geophysical Research: Planets, 125(5)*, e2019JE006264, https://doi.org/10.1029/2019JE006264.

66. **Kerner, H. R.**, Wagstaff, K. L., Bue, B. D., Gray, P., Bell III, J. F., Ben Amor, H (2019). Deep Learning Methods Toward Generalized Change Detection on Planetary Surfaces. *Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(10), pp. 3900-3918, https://doi.org/10.1109/JSTARS.2019.2936771.
67. **Kerner, H. R.**, Ben Amor, H., Bell III, J. F. (2018). Context-Dependent Image Quality Assessment of JPEG-Compressed Mars Science Laboratory Mastcam Images using Convolutional Neural Networks. *Computers and Geosciences*, 118, pp. 109-121, https://doi.org/10.1016/j.cageo.2018.06.001.
68. Kwan, C., Chou, B., Kwan, L., Larkin, J., Ayhan, B., Bell III, J. F., **Kerner, H. R.** (2017). Demosaicing Enhancement using Pixel-Level Fusion. *Signal, Image and Video Processing*, 12(4), pp. 749-756, https://doi.org/10.1007/s11760-017-1216-2.

Peer-Reviewed Conference Workshops (not in proceedings)

69. Corley, I., Lehmann, N., Robinson, C., **Kerner, H.** (2026). torchgeo-bench: An Evaluation Harness for Geospatial Foundation Models. *40th Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2nd Workshop on Advances in Representation Learning for Earth Observation. 🎤 Oral presentation
70. Purohit, M., Gajera, B., Malaviya, V., Mehta, I., Kasodekar, K., Adler, J., Lu, S., Rebbapragada, U., and **Kerner, H.** (2025). Mars-Bench: A Benchmark for Evaluating Foundation Models for Mars Science Tasks. *Machine Learning for the Physical Sciences Workshop at NeurIPS 2025.* 🥇 Best Poster Award
71. Hulkund, N., Maalouf, A., Cai, L., Yang, D., O'Neill, A., Haucke, T., Mukherjee, S., Ramaswamy, V. V., Shen, J. H., Tseng, G., Walmsley, M., Wang, T., **Kerner, H.**, Chen, I. Y., Girdhar, Y., Rus, D., Goldberg, K., Beery, S. (2025). DataS3: Dataset Subset Selection for Specialization. *42nd International Conference on Machine Learning (ICML) Workshops*, DataWorld Workshop, https://openreview.net/forum?id=8F82jNt2uA.
72. Tseng, G., Fuller, A., Reil, M., Herzog, H., Beukema, P., Bastani, F., Green, J. R., Shelhamer, E., **Kerner, H.**, and Rolnick, D. (2025). Galileo: Learning Global & Local Features of Many Remote Sensing Modalities. In *42nd International Conference on Machine Learning (ICML) Workshops*, Terrabytes Workshop, https://openreview.net/forum?id=a2Qn7lTYWL.
73. Purohit, M., Muhawenayo, G., Rolf, E., and **Kerner, H.** (2025). How Does the Spatial Distribution of Pre-training Data Affect Geospatial Foundation Models? *AAAI Workshop on Preparing Good Data for Generative AI: Challenges and Approaches*, https://arxiv.org/pdf/2501.12535.
74. Tseng, G., Zvonkov, I., Purohit, M., Rolnick, D., and **Kerner, H.** (2023). Lightweight, Pre-trained Transformers for Remote Sensing Timeseries. *Neural Information Processing Systems (NeurIPS)*, Climate Change AI Workshop, https://www.climatechange.ai/papers/neurips2023/58.
75. **Kerner, H.**, Sundar, S., and Satish, M. (2023). Multi-Region Transfer Learning for Segmentation of Crop Field Boundaries in Satellite Images with Limited Labels. In *AAAI Conference on Artificial Intelligence Workshops*, https://arxiv.org/abs/2404.00179.
76. Nakalembe, C. L. and **Kerner, H. R.** (2022). Applications and Considerations for AI-EO for Agriculture in Sub-Saharan Africa. In *Proceedings of the Thirty-Sixth AAAI Conference on Artificial Intelligence*, International Workshop on Social Impact of AI for Africa.

77. Tseng, G., **Kerner, H.**, Rolnick, D. (2022). TIML: Task-Informed Meta-Learning for crop type mapping. In *Thirty-Sixth AAAI Conference on Artificial Intelligence*, AI for Agriculture and Food Systems (AIAFS) Workshop.

78. Lacoste, A., Sherwin, E., **Kerner, H.**, Alemohammad, H., Lutjens, B., Irvin, J., Dao, D., Chang, A., Gunturkun, M., Drouin, A., Rodriguez, P., Vazquez, D. (2021). Toward Foundation Models for Earth Monitoring: Proposal for a Climate Change Benchmark. *Neural Information Processing Systems (NeurIPS) Workshops*, Tackling Climate Change with AI, `https://www.climatechange.ai/papers/neurips2021/73/paper.pdf`.

79. Huppertz, R., Nakalembe, C., **Kerner, H.** (2021). Using transfer learning to study burned area dynamics: A case study of Refugee settlements in West Nile, Northern Uganda. *ACM/SIGKIDD Conference on Knowledge Discover and Data Mining (KDD) Workshops*, Humanitarian Mapping, `https://arxiv.org/abs/2107.14372`.

80. Paliyam, M., Nakalembe, C., **Kerner, H.** (2021). Street2Sat: A Machine Learning Pipeline for Generating Ground-truth Geo-referenced Labeled Datasets from Street-Level Images. *International Conference on Machine Learning (ICML) Workshops*, Tackling Climate Change with AI, `https://www.climatechange.ai/papers/icml2021/74.html`.

81. Tseng, G., **Kerner, H.**, Nakalembe, C., and Becker-Reshef, I. (2020). Annual and in-season mapping of cropland at field scale with sparse labels. *Neural Information Processing Systems (NeurIPS) Workshops*, Tackling Climate Change with AI, `https://www.climatechange.ai/papers/neurips2020/29/paper.pdf`.

82. **Kerner, H. R.**, Sahajpal, R., Skakun, S., Becker-Reshef, I., Barker, B., Hosseini, M. (2020). Resilient In-Season Crop Type Classification in Multispectral Satellite Observations using Growth Stage Normalization. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining Workshops*, `https://arxiv.org/abs/2009.10189`.

83. **Kerner, H. R.**, Tseng, G., Becker-Reshef, I., Barker, B., Munshell, B., Paliyam, M., Hosseini, M. (2020). Rapid Response Crop Maps in Data Sparse Regions. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining Workshops*, `https://arxiv.org/abs/2006.16866`. 🎤 Oral presentation

84. **Kerner, H. R.**, Nakalembe, C., Becker-Reshef, I. (2020). Field-Level Crop Type Classification with k-Nearest Neighbors: A Baseline for a New Kenya Smallholder Dataset. *International Conference on Learning Representations (ICLR) Workshops*, `https://arxiv.org/abs/2004.03023`.

Books and Book Chapters

85. Becker-Reshef, I., Bandaru, V., Barker, B., Coutu, S., Deines, J. M., Doorn, B., Eilerts, G., Franch, B., Galvez, A. S., Hosseini, M., Humber, M., Husak, G., Guan, K., Justice, C., Keniston, J., **Kerner, H.**, Mitkish, M., Mobley, K., Munshell, B., Nakalembe, C., Puricelli, E., Sahajpal, R., Skakun, S., Vermote, E., Whitcraft, A., Hansen, M., Salas, B., and Justice, C. (2022). The NASA Harvest Program on Agriculture and Food Security. In: Vadrevu, K.P., Le Toan, T., Ray, S.S., Justice, C. (eds) *Remote Sensing of Agriculture and Land Cover/Land Use Changes in South and Southeast Asian Countries. Springer, Cham.* `https://doi.org/10.1007/978-3-030-92365-5_3`.

86. Aye, K. M., D'Amore, M., Helbert, J., **Kerner, H. R.** (2022). *Machine Learning for Planetary Science*. Elsevier Science and Technology Books, `https://doi.org/10.1016/C2018-0-04220-6`.

87. **Kerner, H.**, Campbell, J., and Strickland, M. (2022). Chapter 1 - Introduction to machine learning. In: Helbert, J., D'Amore, M., Aye, K. M., Kerner, H. R. (eds), *Machine Learning for Planetary Science*, pp. 1-24, https://doi.org/10.1016/B978-0-12-818721-0.00007-0.

Whitepapers and Other Publications

88. Chausse, R., Andersen, D., Howarth, C. N., Mooney, O., Poole, D. N., Raymond, N. A., Kerner, H., Tárano, A. M., Becker-Reshef, I. (2026). The Impact of Systematic Razing of Agricultural Communities around El-Fasher: A Remote Sensing Study. Humanitarian Research Lab at Yale School of Public Health: New Haven, https://files-profile.medicine.yale.edu/documents/0667d53f-af67-415c-99ce-bd9b2f32bc0e.
89. Tseng, G., Zvonkov, I., Purohit, M., Rolnick, D., and **Kerner, H.** (2023). Lightweight, Pre-trained Transformers for Remote Sensing Timeseries. arXiv preprint, https://arxiv.org/abs/2304.14065.
90. Xu, L., Rolf, E., Beery, S., Bennett, J. R., Berger-Wolf, T., Birch, T., Bondi-Kelly, E., Brashares, J., Chapman, M., Corso, A., Davies, A., Garg, N., Gaylard, A., Heilmayr, R., **Kerner, H.**, Klemmer, K., Kumar, V., Mackey, L., Monteleoni, C., Moorcroft, P., Palmer, J., Perrault, A., Thau, D., Tambe, M. (2023). Reflections from the Workshop on AI-Assisted Decision Making for Conservation, *arXiv preprint arXiv:2307.08774*, https://arxiv.org/abs/2307.08774.
91. Nakalembe, C., Justice, C., **Kerner, H.**, Justice, C., and Becker-Reshef, I. (2021). Sowing seeds of food security in Africa, *Eos, 102*, https://doi.org/10.1029/2021EO153329.
92. Azari, A. R., Biersteker, J. B., Dewey, R. M., Doran, G., Forsberg, E., Harris, C. D. K., **Kerner, H. R.**, Skinner, K. A., Smith, A. W. (2020). Integrating Machine Learning for Planetary Science: Perspectives for the Next Decade. *A White Paper to the NRC Planetary Science and Astrobiology Decadal Survey 2023-2032*, https://arxiv.org/pdf/2007.15129.pdf.
93. **Kerner, H. R.** (2019). Machine Learning on Mars: A New Lens on Data from Planetary Exploration Missions. Ph.D. Dissertation, Arizona State University.

Editing

- Aye, K. M., D'Amore, M., Helbert, J., Kerner, H. R. (2022). Machine Learning for Planetary Science. *Elsevier Science and Technology Books.*
- Guest editor, "Monitoring Climate Impacts on Agriculture Using Remote Sensing Techniques," special issue in *Remote Sensing* (2020-2021)
- Guest editor, "Recent Advances in Crop Mapping and Monitoring Using Remote Sensing Data," special issue in *Remote Sensing* (2020-2021)
- Guest editor, "Advances in AI applications for small-scale agricultural systems," special issue in *Frontiers in AI in Food, Agriculture, and Water* (2020-2021)

PUBLIC DATASETS AND OPEN-SOURCE SOFTWARE

2026 Geospatial Evaluation Harness (geo-eval-harness), https://github.com/torchgeo/geo-eval-harness

2026 Global Fields of The World (FTW) 2nd Edition (global agricultural field map for 2017-2025), https://source.coop/ftw/global-data-2e
2026 EarthShift testbed, https://github.com/kerner-lab/earthshift
2026 Fields of The World Global 10m Field Boundaries (global agricultural field map for 2024-2025), https://source.coop/ftw/global-data
2025 GEO-Bench-2, https://github.com/The-AI-Alliance/GEO-Bench-2
2025 Mars-Bench, https://mars-bench.github.io/
2024 Fields of The World benchmark dataset and tools, https://fieldsofthe.world/
2023 GEO-Bench, https://github.com/ServiceNow/geo-bench
2022 OpenMapFlow python package, https://pypi.org/project/openmapflow/
2022 CropHarvest python package, https://pypi.org/project/cropharvest/
2021 DORA python package, https://github.com/nasaharvest/dora
2021 CropHarvest: A global dataset for crop type mapping https://zenodo.org/record/5567762
2021 Kenya 10m Cropland Map and Labels (2019) https://zenodo.org/record/4271144#.YIAzqpNKhFc
2020 MSL Curiosity Rover Images with Science and Engineering Classes https://doi.org/10.5281/zenodo.3892023
2020 Togo 10m Cropland Map and Labels (2019) https://doi.org/10.5281/zenodo.3836628
2020 Mars Novelty Detection Mastcam Labeled Dataset https://doi.org/10.5281/zenodo.1486195
2019 Dynamic Albedo of Neutrons (DAN) Simulated and Observed Die-Away Data https://doi.org/10.5281/zenodo.3592014
2019 Planetary Surface Features Change Detection Dataset https://doi.org/10.5281/zenodo.2373797

SELECTED INVITED TALKS

2026 "The Promise and Pitfalls of Geospatial AI for Food Security." Brigham Young University (BYU) AI for Social Innovation Conference, Provo Canyon.
2026 "Multimodal foundation models in remote sensing: domain-specific challenges and solutions." Modeling the World Through an Imperfect Lens (MONTI) Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Denver.
2026 "Are geospatial foundation models ready for science applications?" CV4Science Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Denver.
2026 "Multimodal Pretraining Under Sensor Diversity: Lessons Learned from Foundation Models for Earth and Mars." GeoCV Workshop, Winter Conference on Applications of Computer Vision (WACV), Tucson.
2026 "Multimodal Pretraining Under Sensor Diversity: Lessons Learned from Foundation Models for Earth and Mars." Alphabet Modeling Talks, Virtual.
2026 "Multimodal Pretraining Under Sensor Diversity: Lessons Learned from Foundation Models for Earth and Mars." Learning | Information | Optimization | Networks | Statistics (LIONS) Seminar, Arizona State University, Tempe.
2025 "Satellite Machine Learning for Agriculture and Beyond." USDA Geospatial Data Sharing Seminar, Virtual.
2025 "Encoding the planet: foundation models for Earth observation." NOAA AI Workshop, Boulder.

2025 "Machine Learning Foundation Models for Satellite Remote Sensing and Agriculture." USDA Agricultural Research Service (ARS) AI Seminar Series, Virtual.

2025 "Towards planetary intelligence: machine learning for remote sensing data that creates impact for science and society." Plenary Keynote, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Tucson.

2025 "Geospatial foundation models for the people." Computer Vision for Earth Observation (CV4EO) Workshop, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Tucson.

2025 "Fields of The World (FTW!): A New Machine Learning Dataset for Agricultural Field Boundary Segmentation on Four Continents." ESIP Winter Meeting, Virtual.

2025 "Foundation models for Earth and space: The key to AI applications, or distracting hype?" Earth-Life Science Institute (ELSI) 13th Symposium, Tokyo, Japan.

2024 "Foundation models for Earth and space: The key to AI applications, or distracting hype?" Science Understanding through Data Science Seminar, NASA Jet Propulsion Laboratory, Pasadena.

2024 "How can we create geospatial foundation models that are useful for Earth impact?" Computer Vision for Ecology Workshop, European Conference on Computer Vision (ECCV), Milan.

2024 "Harnessing Machine Learning and Satellite Data for Planetary-Scale Impact." Department of Astronomy and Planetary Science Colloquium, Northern Arizona University, Flagstaff.

2024 "Harnessing Machine Learning and Satellite Data for Planetary-Scale Impact." United Nations and ITU AI for Good, Online.

2024 "Foundation Models in the Real World (literally)." International Conference on Machine Learning (ICML) Workshop on Foundation Models in the Wild, Vienna.

2024 "Unlocking the Potential of Planetary-Scale Machine Learning for a Sustainable Future." Keynote, ISPRS Technical Symposium II, Las Vegas.

2024 "Unlocking the Potential of Planetary-Scale Machine Learning for a Sustainable Future." Spotlight, Machine Learning and Data Management for Earth Observation Workshop, BIFOLD (Berlin Institute for the Foundations of Learning and Data), Berlin.

2023 "Unlocking the Potential of Planetary-Scale Machine Learning for a Sustainable Future." Keynote, NeurIPS 2023, Sustainable Computing Workshop, New Orleans.

2023 "Unlocking Geospatial Machine Learning for Global Agricultural Monitoring." Keynote, DA3 Symposium, Kansas State University, Manhattan.

2023 "Unlocking the Potential of Planetary-Scale Machine Learning for a Sustainable Future." Computer Science and Engineering Department Colloquium, Washington University St. Louis, St. Louis.

2023 "AI for the Digital Planet." NSF AI Planning Institute Seminar, Carnegie Mellon University, Pittsburgh.

2023 "AI for the Digital Planet(s)." School of Earth and Space Exploration Colloquium, Arizona State University, Tempe.

2022 "Supporting Food Security in Africa using Machine Learning and Earth Observations." Machine Learning and the Physical Sciences Workshop, Neural Information Processing Systems (NeurIPS) 2022, New Orleans/Virtual.

2022 "AI and Earth Observations for Global Agricultural Monitoring and Food Security." AI Helps Ukraine fundraiser conference, Virtual.

2022 "Supporting Food Security in Africa using Machine Learning and Earth Observations." Computer Vision for Ecology Summer School, Caltech/Virtual.

2022 "Street2Sat: turning roadside images into ground-truth labeled datasets for machine learning." AgricultureVision workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans.

2022 "AI for Earth Observations and Food Security—going beyond test metrics." AI4FoodSecurity Awards Ceremony (keynote), Virtual.

2022 "Advancing Global Food Security and Sustainable Development with ML and Earth Observations." Remote Sensing Interdisciplinary Graduate Education Program (IGEP) Seminar, Virginia Tech, Blacksburg, VA.

2022 "Advancing Global Food Security and Sustainable Development with ML and Earth Observations." AI Seminar, Oregon State University, Virtual.

2021 "Advancing Global Food Security and SDGs with Machine Learning and Earth Observations." Computer Science Department Seminar, University of Maryland, College Park, MD.

2021 "Advancing Global Food Security and SDGs with Machine Learning and Earth Observations." Forward Summit, Puerto Rico Science Trust, Virtual.

2021 "Advancing Global Food Security and SDGs with Machine Learning and Earth Observations." iCube Institute Seminar, Strasbourg, France.

2021 "Advancing Global Food Security and SDGs with Machine Learning and Earth Observations." NASA Marshall Space Flight Center, IMPACT Tech Talk, Virtual.

2021 "Novelty-guided onboard targeting and tactical planning for Mars rovers." Applications of Statistical Methods and Machine Learning in the Space Sciences, Space Science Institute, Virtual.

2021 "Enhancing Global Food Security with Machine Learning and Planet Data." Planet Colloquium, Virtual.

2021 "Mars, Machine Learning, and the Search for Life beyond Earth: How the Mars Perseverance and Curiosity rovers can use machine learning to detect the unknown." Ubiquity Ventures Public Event, Virtual.

2021 "Enhancing Global Food Security with Machine Learning and Earth Observations." World Resources Institute AI for Impact Series, Virtual.

2021 "The Power of ML and EO to Enhance Global Food Security." ESIP Winter Meeting, Virtual.

2020 "Eyes in the sky without boots on the ground: Using satellites and machine learning to monitor agriculture and food security during COVID-19." NeurIPS Workshop on AI for Earth Science, Virtual.

2020 "How to Shape Your Career." IEEE GRSS and IGARSS Webinar Series.

2020 "Using Machine Learning in Space." Ubiquity Extended Team Webinar, Virtual.

2020 "Monitoring Agriculture at the Field Scale using Satellite Data and Machine Learning." Measuring Development 2020: Data Integration and Data Fusion, Virtual.

2020 "Enhancing Planetary Exploration Mission Planning and Data Analysis using Machine Learning." Solar System Exploration Division Seminar, NASA Goddard Space Flight Center, Greenbelt, MD.

2020 "Machine Learning for Agricultural Monitoring." Advancing Application of Machine Learning Tools for NASA's Earth Observation Data, Washington, DC.

2019 "Actionable Insights from Remote Sensing Enabled by Machine Learning, from Earth to Mars." International Space University, Strasbourg, France.

2019 "Actionable Insights from Remote Sensing Enabled by Machine Learning, from Earth to Mars." Arizona State University, Tempe, AZ.

2019 "Actionable Insights from Remote Sensing Enabled by Machine Learning, from Earth to

Mars." Women in Data Science at Stanford Earth, Palo Alto, CA.

2019 "Machine Learning for Remote Sensing." Committee on Seismology and Geodynamics (COSG) Fall Meeting, National Academies of Science, Engineering, and Medicine, Washington, DC.

2019 "AI and Machine Learning." Space4Earth Hackathon, 70th International Astronautical Congress, Washington, DC.

CONFERENCE SERVICE

2026 Session Chair/Co-Convener, "Transfer Learning in Remote Sensing: Overcoming Data Scarcity and Domain Shift," American Geophysical Union (AGU) Fall Meeting, San Francisco, Dec 7-11.

2026 Session Chair/Co-Convener, "Machine Learning and Data Science Methods in Planetary Science," American Geophysical Union (AGU) Fall Meeting, San Francisco, Dec 7-11.

2025 Session Chair/Co-Convener, "Machine Learning and Data Science Methods in Planetary Science," American Geophysical Union (AGU) Fall Meeting, New Orleans, Dec 15-19.

2025 Session Chair/Co-Convener, "Advancing Artificial Intelligence for Remote Sensing: Overcoming Data Scarcity and Domain Shift," American Geophysical Union (AGU) Fall Meeting, New Orleans, Dec 15-19.

2026 Co-organizer, EarthVision Workshop at Computer Vision and Pattern Recognition (CVPR) 2026, Denver.

2026 Co-organizer, 4th Machine Learning for Remote Sensing Workshop at International Conference on Learning Representations (ICLR) 2026, Rio de Janeiro, April 27.

2025 Co-organizer, EarthVision Workshop at Computer Vision and Pattern Recognition (CVPR) 2025, Nashville, June 11.

2025 Lead Organizer, 3rd Machine Learning for Remote Sensing Workshop at International Conference on Learning Representations (ICLR) 2025, Singapore, April 27.

2024 Session Chair/Co-Convener, "Machine Learning and Data Science Methods for Planetary Science," American Geophysical Union (AGU) Fall Meeting, Washington DC, Dec 9-13.

2024 Co-organizer, 2nd Machine Learning for Remote Sensing Workshop at International Conference on Learning Representations (ICLR) 2024, Vienna, May 11.

2024 Co-organizer, EarthVision Workshop at Computer Vision and Pattern Recognition (CVPR) 2024, Seattle, June 17.

2023 Session Chair/Co-Convener, "Highlighting Solution-Based Science: Connecting with Local Partners for Action," American Geophysical Union (AGU) Fall Meeting, San Francisco, December 11-15.

2023 Session Chair/Co-Convener, "Beyond the Black Box: advancing geo-ML by incorporating context with specialized architectures, benchmark datasets, and tailored notions of interpretability," American Geophysical Union (AGU) Fall Meeting, San Francisco, December 11-15.

2023 Session Chair/Co-Convener, "Machine Learning and Data Science Methods for Planetary Science," American Geophysical Union (AGU) Fall Meeting, San Francisco, Dec 11-15.

2023 Session Chair/Co-Convener, "Applications of Earth Observations for United States Agriculture," American Geophysical Union (AGU) Fall Meeting, San Francisco, December 11-15.

2023 Co-organizer, EarthVision Workshop at Computer Vision and Pattern Recognition (CVPR) 2023, Vancouver, June 18.

2023 Lead Organizer, Machine Learning for Remote Sensing Workshop at International

Conference on Learning Representations (ICLR) 2023, Kigali, May 1-5.

2022 Session Chair/Co-Convener, "Machine Learning for Planetary Science," American Geophysical Union (AGU) Fall Meeting, Chicago, December 12-16.

2022 Organizer/Speaker, Machine Learning for Remote Sensing and Applications in Agriculture and Food Security tutorial (invited) at Computer Vision and Pattern Recognition (CVPR) 2022, New Orleans, June 19-24.

2022 Co-organizer, "3rd International Workshop and Prize Challenge on Agriculture-Vision: Challenges & Opportunities for Computer Vision in Agriculture," Computer Vision and Pattern Recognition (CVPR) 2022, New Orleans, June 19-24.

2021 Organizing Committee Member, "On the Pathway to a Digital Earth", Joint AI/ML Workshop between UMD and NASA Goddard Space Flight Center, College Park, September 22.

2021 Session Chair/Co-Convener, "Machine Learning for Planetary Science," American Geophysical Union (AGU) Fall Meeting, New Orleans and Online, December 13-17.

2021 Program Committee, Humanitarian Mapping Workshop, KDD, August 14-18.

2021 Organizer/Session Chair, The Trillion Pixel Challenge Workshop, Virtual (hosted by ORNL), April 21-22.

2020 Session Chair/Co-Convener, "Machine Learning for Planetary Science," American Geophysical Union (AGU) Fall Meeting, Virtual Conference, December 1-17.

2020 Co-Chair, "Robots in the Wild: Challenges in Deploying Robust Autonomy for Robotic Exploration," Workshop at Robotics: Science and Systems (RSS), Virtual Conference, July 12.

2019 Session Chair/Co-Convener, "Machine Learning for Planetary Science," American Geophysical Union (AGU) Fall Meeting, San Francisco, CA, December 9-13.

2018 Session Co-Convener, "Machine Learning in Planetary Science: Introductions and Applications," American Geophysical Union (AGU) Fall Meeting, Washington, DC, December 10-14.

2017 Session Co-Convener, "Rise of Machine Learning: Salvation for Planetary Science in Times of Increasing Data Volume and Complexity," American Geophysical Union (AGU) Fall Meeting, New Orleans, LA, December 11-15.

2017 Co-Chair, NewSpace Europe Conference, Luxembourg City, November 16-17.

TEACHING EXPERIENCE

Course Coordinator
Foundations of Machine Learning (CSE 475), Arizona State University

Courses Taught
Arizona State University, School of Computing and Augmented Intelligence

- **CSE 598** – Special Topics: Machine Learning for Remote Sensing — Spring 2024
- **CSE 475** – Foundations of Machine Learning — Fall 2023/24/26; Spring 2025/26
- **CSE 572** – Data Mining — Fall 2022; Spring 2023/25

University of Maryland, Department of Geographical Sciences

- **GEOG 670** – Open Source GIS — Spring 2022

University of North Carolina at Chapel Hill, Department of Computer Science

- CS for People Who Don't Know CS (Yet!) Spring 2015

Tutorials and Capacity Building

Apr 2026	**Cropland Mapping using Geospatial Embeddings**, Machine Learning for Remote Sensing Workshop, ICLR 2026, Rio de Janeiro (`https://arxiv.org/abs/2511.02923`)
Apr 2026	**Getting Started with OlmoEarth: From Embeddings to Fine-tuning**, Machine Learning for Remote Sensing Workshop, ICLR 2026, Rio de Janeiro (`https://openreview.net/pdf?id=IAyXlDwCYP`)
Apr 2026	**Fields of The World: A Field Guide for Extracting Agricultural Field Boundaries**, Machine Learning for Remote Sensing Workshop, ICLR 2026, Rio de Janeiro (`https://arxiv.org/abs/2602.08131`) 🏅 Best Tutorial Award
Dec 2025	**Agricultural Monitoring with Fields of The World (FTW)**, Climate Change AI Workshop, NeurIPS 2025, San Diego
June 2024	**Cropland Mapping Training for NASA SERVIR West Africa Hub** (Centre de Suivi Ecologique), Dakar, Senegal
May 2023	**Machine Learning for Remote Sensing tutorial**, hosted at Carnegie Mellon University Africa (Kigali, Rwanda) and led by ASU, NASA Harvest, and CMU
Feb 2023	**'Āina Data Stewards program kickoff**, hosted at University of Hawaii Maui College and led by ASU, UMD, and NASA Harvest/ACRES consortium team members
Sep 2022	**Argentina Multilateral training program led by NASA Harvest**, the Buenos Aires Grain Exchange (Bolsa de Cereales), the Argentinian Ministry of Education, and the Inter-American Development Bank
Aug 2022	**Scalable Cropland Mapping Workshop**, hosted at University of Maryland for Regional Centre on Mapping Resources and Development (RCMRD)
Jun 2022	**Machine Learning for Remote Sensing and Applications in Agriculture and Food Security tutorial (invited)** at Computer Vision and Pattern Recognition (CVPR)
Mar 2022	**AI and Earth Observations for Agriculture tutorials** led by UMD and NASA Harvest at Regional Centre on Mapping Resources and Development (RCMRD) and Rwanda Space Agency (RSA)
Sep 2021	**Crop Yield Modeling Workshop**, Regional Centre for Mapping of Resources for Development (RCMRD), NASA Harvest and NASA SERVIR (PI: Nakalembe)
Aug–Dec 2021, 2022	**Course on remote sensing and ML technologies for agriculture**, Argentina Rural Schools Education Program led by NASA Harvest and Bolsa de Cereales

MEDIA

Opinion Articles

1. Sowing Seeds of Food Security in Africa. *Eos*, 2021.
2. Too many AI researchers think real-world problems are not relevant. *MIT Technology Review*, 2020.
3. Our path to Mars needs to look beyond launch. *Houston Chronicle*, 2016.
4. Space technology can help sustain Earth. *Scientific American*, 2016.

5. What's The Point? The Real Reason Scientists Study Space. *Space.com*, 2015.
6. The Space Destination Debate Gets Us Nowhere... Literally. *Space.com*, 2015.
7. It's Not Them It's You: Why Top Tech Talent Isn't Going to the Satellite Industry. *Via Satellite*, 2015.

Featured Articles, Podcasts, and Other Media

8. Building "large Earth models" with Hannah Kerner, Lab Coat Optional, 2026.
9. ASU researcher's AI work earns US nomination for international science prize, ASU News, 2026.
10. AI is watching for wildfires across the drought-stricken West, NPR/WBUR, 2026.
11. Mapping South America and Beyond with Fields of The World V2, Satellite Image Deep Learning Podcast, 2026.
12. 12 million images later, Mars starts to make sense, ASU News, 2026.
13. Ontario Tech panel examines AI's environmental risks and opportunities, Ontario Tech University News, 2026.
14. ASU's Hannah Kerner recognized among top global AI innovators, ASU News, 2025.
15. Monitoring Crops from Space: How NASA Harvest Tracks Agriculture in Conflict Zones Like Tigray, Ethiopia, NASA Harvest Blog, 2025.
16. A career in orbit, ASU News, 2025.
17. Finding the fields of the world, ASU News, 2025.
18. Presto: A Smarter Way to Use Satellite Data to Help Farmers, NASA Harvest Blog, 2025.
19. Harvesting satellite insights for Maui County farmers, ASU News, 2024.
20. Innovation Bridge Community Spotlight: Dr. Hannah Kerner, Taylor Geospatial Engine Blog, 2024.
21. Emerging machine learning expert leads Kenya AI workshop, ASU Full Circle, 2024.
22. With help from AI, ASU researcher develops models to address climate change, other global issues, ASU News, 2024.
23. NASA is helping Hawai'i farmers grow more food with AI, Agriculture Dive, 2023.
24. Maui United Way and NASA Collaboration Expedites Aid for Maui Wildfire Survivors, Maui Times, 2023.
25. ASU researcher combats food insecurity with AI, ASU News, 2023.
26. NASA Harvest Expands Food Security Work In Maui County, HI With Community-Based Partners, NASA Harvest Blog, 2022.
27. AskSME: Dr. Hannah Kerner - Artificial Intelligence Lead, Close-up with a NASA Subject Matter Expert, NASA eClips Video, 2021.
28. Real World: Food Security - Monitoring Crops from Space, NASA eClips Video, 2021.
29. Remote Sensing with ML and Starting Your Professorship, How to Do Grad School, 2021.
30. Cracking the Code for Healthy Crops with Hannah Kerner, NASA Applied Sciences Profile, 2021.
31. Hannah Kerner on Remote Sensing with Machine Learning, Time Horizons Podcast, 2021.

32. Space Scientists Naia Butler-Craig and Hannah Kerner Make Forbes' Under 30 List, Via Satellite, 2020.
33. How Farmers Can Help NASA Assess the Impacts of the Derecho in Iowa, AgriTalk Radio, 2020.
34. NASA Funds Eight New Projects Exploring Connections Between the Environment and COVID-19. *NASA Earth Sciences Division*, 2020.
35. Smart Machines: Enabling a New Era of Planetary Exploration. *CuttingEdge*, 2020.
36. Harvest Hub: Food Security from Space. *Via Satellite*, On Orbit podcast, 2020.

SERVICE

Reviewing / Program Committees

Recognition

- Top Area Chair, NeurIPS 2026
- Outstanding Reviewer, CVPR 2025 (711 reviewers recognized out of 12,593, or 5.6%)

Conference Peer Review

- International Conference on Learning Representations (ICLR)
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
- International Conference on Computer Vision (ICCV)
- International Conference on Machine Learning (ICML)
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- Association for the Advancement of Artificial Intelligence (AAAI)
- Neural Information Processing Systems (NeurIPS)

Conference Area Chair, Meta-Reviewer, or Senior Program Committee Member

- Neural Information Processing Systems (NeurIPS)
- International Conference on Machine Learning (ICML)
- Association for the Advancement of Artificial Intelligence (AAAI)
- International Conference on Learning Representations (ICLR)

Workshop Peer Review and Meta-Review

- Workshop on Machine Vision for Earth Observation and Environment Monitoring (BMVC)
- Humanitarian Mapping Workshop (KDD)
- Climate Change AI Workshop (ICML, ICLR)
- Women in Machine Learning (NeurIPS)

Journal Peer Review

- *ISPRS Journal of Photogrammetry and Remote Sensing*
- *Earth System Science Data (ESSD)*

- *Nature Communications*
- *Nature Food*
- *Science Robotics*
- *Precision Agriculture*
- *Remote Sensing of Environment*
- *Journal of Selected Topics in Applied Earth Observations and Remote Sensing*
- *Frontiers in Sustainable Food Systems*
- *Hydrology and Earth System Science*
- *Agronomy*
- *Agriculture*
- *Remote Sensing*
- *IEEE Transactions on Geoscience and Remote Sensing*

Funding/Sponsored Research Peer Review

- NSF CISE Directorate: III Core Medium
- Climate Change AI Innovation Grants
- NASA ROSES
- Puerto Rico Science Trust

Organizations, Advisory Groups, and Committees

Internal

2025-	AI Graduate Program Committee (ASU School of Computing and Augmented Intelligence)
2022-2023	Graduate Admissions Committee (ASU School of Computing and Augmented Intelligence)
2021	Unlearning Racism in the Geosciences (URGE) Pod (University of Maryland)
2020-2022	Diversity, Equity, Inclusion, and Anti-Racism Committee (University of Maryland)
2020-2022	Graduate Committee (UMD Department of Geographical Sciences)
2020-2021	Department Committee Voting Representative (UMD Department of Geographical Sciences)

External

2026-	NASA Lifelines Food Security Working Group
2025-	Academic Advisory Board, Forecasting Research Institute Longitudinal Expert AI Panel (
2024-	Rescue Sight Board of Advisors
2024-2026	Data Archiving and Access Requirements Working Group (DAARWG) NOAA Science Advisory Board (SAB), Federal Advisory Committee
2024	Pu'uhonua Kauluwehi Advisory Council, UH Maui College
2020-2021	Technical Advisory Panel, The Lacuna Fund: Our Voice on Data Meridian Institute and Rockefeller Foundation
2020-	Co-organizer, Machine Learning for Remote Sensing Online Discussion Group, `https://bit.ly/2KoEX7K`
2020	Technical Committee, 2020 NSF CPS Challenge "SoilScope – Mars edition"
2019-2020	Volunteer, Board of Directors, Research & Policy Committee Women in Machine Learning (WiML)
2019-	Member (advising early-stage investments) Ubiquity Ventures Extended Team (UXT)
2015-	Member, Board of Advisors Students for the Exploration and Development of Space (SEDS) USA
2018-2019	Co-Chair, Women in Science Program School of Earth and Space Exploration, Arizona State University
2015-2019	Member, Colloquium Committee School of Earth and Space Exploration, Arizona State University

Advising and Mentoring

PhD Students – Thesis committee chair or co-chair

1. Sarah Peterson (ASU Exploration Systems Design, PhD), 2024-2028
2. Amandeep Kaur (ASU Computer Science, PhD), 2024-2029
3. Bhanu Tokas (ASU Computer Science, PhD), 2024-2028
4. Chenwei Cui (ASU Computer Science, PhD), 2023-2028
5. Gedeon Muhawenayo (ASU Computer Science, PhD), 2023-2028
6. Rahul Nair (ASU Computer Science, PhD), 2022-2027
7. Mirali Purohit (ASU Computer Science, PhD), 2022-2026

PhD Students – Thesis committee member (🎓: Graduated)

8. Jisoo Lee (ASU Computer Science, PhD), 2023-2028
9. Protik Bose Pranto (ASU Computer Science, PhD), 2022-2027
10. Sizhe Wang (ASU Computer Science, PhD)
11. Jianghai Peng (ASU Geological Sciences, PhD), 2022-2027
12. Zifan Yu (ASU Computer Science, PhD), 2020-2026
13. Hyunho Lee (ASU Geographic Information Systems, PhD), 2022-2026 🎓
14. Gabriel Tseng (McGill University/Mila, Computer Science, PhD), 2020-2025 🎓
15. Paul Horton (ASU Exploration Systems Design, PhD), 2019-2025 🎓
16. Laurel Hopkins (Oregon State University, Computer Science, PhD), 2018-2024 🎓

M.S. Students – Thesis committee chair, co-chair, or member (🎓: Graduated)

1. Kalli Harshman (ASU Coastal and Marine Science and Management, MS), 2024-Present
2. Bradley Peterson (ASU Computer Science, MS), 2024-Present
3. Benjamin Herrera (ASU Computer Science, MS), 2025-2026 🥇 NSF GRFP 🎓
4. Bhanu Tokas (ASU Computer Science, MS), 2022-2024 🎓
5. Matthew Watson (ASU Computer Science, MS), 2023 🎓
6. Ujjwala Anantheswaran (ASU Computer Science, MS), 2022-2023 🎓
7. Ivan Zvonkov (University of Maryland, Computer Science, MS), 2021-2023 🎓

M.S. Students – Project advisor

8. Hunter Blevins (ASU Geography, MA), 2025-2026
9. Bimal Gajera (ASU Computer Science, MS), 2024-2025
10. Vatsal Malaviya (ASU Computer Science, MS), 2024-2025
11. Snehal Chaudhari (ASU Computer Science, MS), 2023-2024
12. Kunal Kasodekar (ASU Computer Science, MS), 2023-2024
13. Kartik Jawanjal (ASU Computer Science, MS), 2023-2024
14. Aninda Ghosh (ASU Robotics, MS), 2023-2024
15. Manthan Satish (ASU Robotics, MS), 2022-2023
16. Keun Park (ASU Computer Science, MS), 2022-2023
17. Aditya Mohan (ASU Computer Science, MS), 2022-2024
18. Aditya Shrivastava (ASU Computer Science, MS), 2022
19. Afolarin Lawal (UMD Geographical Sciences, MS), 2020-2021

Undergraduate Students – Thesis committee chair, co-chair, or member

1. Michael Cencimino (ASU Pre-Law, BS, Barrett Honors thesis), 2025
2. Nicholas Johnson (ASU Computer Science, BS, Barrett Honors thesis), 2024
3. Anant Rastogi (ASU Computer Science, BS, Barrett Honors thesis), 2024
4. Rini Jain (ASU Computer Science, BS, Barrett Honors thesis), 2024
5. Sloan Cooney (ASU Computer Science, BS, Barrett Honors thesis), 2024
6. Saisumana Konatam (ASU Computer Science, BS, Barrett Honors thesis), 2023

Undergraduate Students – Project advisor

7. Catlynh Nguyen (ASU Computer Science, BS), 2025-Present
8. Benjamin Herrera (ASU Computer Science, BS), 2024-2025
9. Adam Yang (UMD Computer Science, BS), 2022-2023
10. Ekenedilichukwu Ndu (UMD Computer Science, BS), 2022
11. Maryann Vazhapilly (UMD Computer Science, BS), 2022

12. Arushi Patel (UMD Computer Science, BS), 2022
13. Sophia Owens (UMD Geographical Sciences, BS), 2022
14. Logan Daytner (UMD Geographical Sciences, BS), 2022
15. Chin-Yun Kuei (UMD Geographical Sciences, BS), 2022
16. Eva Utzschneider (UMD Computer Science, BS), 2021-2022
17. Madhava Paliyam (UMD Computer Science, BS), 2020-2022
18. Kevin Liu (UMD Computer Science, BS), 2021
19. Bryce Dubayah (UMD Computer Science, BS), 2021
20. Sakshum Kulshrestha (UMD Computer Science, BS), 2021
21. Yao Poudima (UMD iSchool), 2021
22. William Cao (UMD Computer Science, BS), 2020-2021
23. Bissaka Kenah (UMD Computer Science, BS), 2020-2021
24. Avi Grant (UMD Geographical Sciences, BS), 2020-2021
25. Favour Nerrise (UMD Computer Science, BS), 2020-2021

High School Students – Project advisor

1. Vedant Janapaty (Silver Creek High School), 2023-2024
2. Saketh Sundar (River Hill High School), 2021-2023
3. Vinay Raman (Montgomery Blair High School), 2021-2022
4. Dhruv Pai (Montgomery Blair High School), 2020-2022

Postdoctoral Researchers – Advisor or co-advisor

1. Savannah Ferretti, Postdoctoral Researcher (Arizona State University), 2026-Present
2. Kelsey Doerksen, Postdoctoral Researcher (Arizona State University), 2025-Present
3. Nils Prieur, Marie Skłodowska-Curie Postdoctoral Fellow (Stanford, University of Oslo), 2021-2024

Other advisees, staff, or research team members

1. Katie Roehrick, Program Manager, Arizona State University, 2026-Present
2. Hunter Blevins, Data Scientist, Arizona State University, 2026-Present
3. Ana M. Tárano, Assistant Research Professor, Arizona State University, 2024-Present
4. Ivan Zvonkov, Faculty Specialist, University of Maryland, 2022-2025
5. Benjamin Yeh, Intern, University of Maryland, 2023
6. Adebowale Adebayo, Intern, University of Maryland, 2022-2023
7. Abena Boatemaa Asare-Ansah, Intern, University of Maryland, 2022-2023

Professional Society Membership
Member, Institute of Electrical and Electronics Engineers (IEEE)
Member, Computer Vision Foundation (CVF)
Member, Association for the Advancement of Artificial Intelligence (AAAI)
Member, American Geophysical Union (AGU)